

# Youbin He

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## Education

 **The Hong Kong Polytechnic University**

Hong Kong | Sep. 2025 – Jun. 2029 (Expected)

 **BEng (Hons) in Aviation Engineering, minor in Artificial Intelligence & Data Analytics (AIDA)**

## Publications

**Youbin He, Siwei Wang. P-SRM: Selective Recovery of Rejected Predictions in Visual Tracking.**

Manuscript submitted to ICASSP 2027; under review.

[arXiv](#) | [GitHub](#)

## Projects

**P-SRM: Lightweight Post-Rejection Selective Recovery for Visual Tracking**

Research project

- **Problem formulation:** While reproducing TrackNetV3, TAPIR, KCF, and other trackers, identified correctly localized candidates discarded by native rejection. Proposed P-SRM through candidate-error categorization and analysis of high-scoring false acceptances.
- **Method design:** Built a unified evidence adapter encoding response maps, candidate support, and valid regions as  $3 \times 32 \times 32$  tensors. Combined these with candidate geometry in a CNN with approximately 1.23M parameters, using auxiliary target-presence supervision, pAUC optimization, native rejection margins, and motion history for selective recovery while preserving native accepted outputs and candidate coordinates.
- **Evaluation:** Ran three-seed experiments across six methods, four datasets, and nine configurations spanning three tracking tasks. Achieved maximum gains of 1.27 percentage points in pooled F1 and 32.13 points in  $AP_r$  over native baselines; assessed component contributions through ablations and additional inference cost through timing measurements.

**HomeSafe: Action Safety Analysis for Household Robots under Limited Perception**

Personal project

- **Problem formulation:** Modeled household robot action safety under limited perception as three-way authorization: allow, deny, or insufficient evidence. Paired sensing configurations with decision rules to assess dangerous authorizations, excessive stopping, and unresolved decisions.
- **Method design:** Restored BEHAVIOR-1K scenes in NVIDIA Isaac Sim through OmniGibson; built a static Unitree Go2 model from official URDF and digital assets. Configured transforms, RGB and linear-depth capture, and ray-based 2D ranging; implemented swept-region Action-ROI projection and YOLO-World detection, with independent collision queries for reference judgments.
- **Data Collection & results:** Collected multimodal observations from 80 development states, producing 21,501 detections and 400 state-action records. Visualization and projection analysis revealed near-field coverage gaps and camera-extrinsic errors. Revised the camera to a  $120^\circ$  field of view and  $40^\circ$  downward tilt, updated and tested the sensing and projection code, and generated multimodal scene reports.

## Experience

 **Lens Intelligent Robot (Changsha) Co., Ltd. — Intern**

Changsha | Jun. – Sep. 2026

- **Design & validation:** Contributed to 6-DOF hand development and design input for the ongoing 22-DOF hand. Used motion analysis, interference checks, and iterative 3D printing to refine structures, removing approximately  $20.31 \text{ mm}^3$  of interference. Assembled prototypes, tested homing and binding, and supported motor and embedded control-software validation.
- **Manufacturing & debugging:** Supported BrainCo 13/16-DOF hand manufacturing; organized BOMs and SAP material records for eight configurations and redrew more than 40 engineering drawings. Delivered assembly training, supported validation of six prototype pairs, and assisted with RS-485 configuration, cable soldering, and test-software debugging.
- **Embodied AI deployment & evaluation:** Deployed Gemini 2 depth cameras on humanoid and quadruped robots and integrated OpenVLA. Completed end-to-end training and testing for tasks spanning locomotion and object grasping, and analyzed results across seven scene categories in the test environment.
- **Client demonstration:** Configured the SDK and PySide6 interface for a 21-DOF hand demonstration to a visiting NVIDIA team. Resolved serial-port and connection issues and explained DOF configuration, tactile sensing, and MIT impedance-control-based finger actuation in English.

## Skills

**Programming & Machine Learning:** Python, MATLAB, Isaac Sim, TypeScript, PyTorch, NumPy, OpenCV, Git, Shell, Linux/WSL2, Conda environments.

**Mechanical Design & Hardware:** SolidWorks, Creo, part and board-outline drawings, 3D-printed prototyping, assembly testing.

**Languages:** Chinese (native); English (IELTS 6.5, 2024).